The Chiral Anomaly in Condensed Matter Systems

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Abstract: In this work, the topological nature of Dirac and Weyl semimetals (DSMs and WSMs), as well as the chiral anomaly, is outlined using the Berry phase and Atiyah-Singer index theorem. The use of angle-resolved photoemission spectroscopy in the search for DSMs and WSMs is discussed, and the most promising materials are identified. Furthermore, the effect of the chiral anomaly on electron transport in WSMs is deduced, namely, a negative linear magnetoresistance (LMR). Lastly, the search for the chiral anomaly in WSMs via anomaly-induced negative LMR is reviewed, and the experimental hurdles introduced by current jetting artefacts are discussed.

Contents

1 Introduction

The concept of quasiparticle dynamics has been an invaluable tool used by condensed matter physicists to study phenomena in quantum materials since the creation of Landau's Fermi-liquid theory [1]. Thus, whenever elementary particle interactions are discovered, condensed matter physicists are often nearby, searching for their quasiparticle ramifications in novel quantum materials. This work reviews the search for Dirac and Weyl semimetals which possess massless Dirac and Weyl fermions as their quasiparticle excitations [2]. These systems are of great interest in solid-state physics. In particular, one of the most exotic results in quantum field theory, the chiral anomaly [3, 4], is predicted to manifest in these materials [5]. We will examine how Dirac and Weyl semimetals are topological materials, because of their topologically nontrivial electronic band structures, before outlining the characterisation of the chiral anomaly as a topological invariant of quantum electrodynamics using the Atiyah-Singer index theorem [6]. Finally, we will deduce the effect of the chiral anomaly on magnetoresistance in Weyl semimetals [5] and review recent efforts to observe this effect.

2 Background

2.1 Relativistic quantum mechanics

In quantum mechanics, the Schrodinger equation is not Lorentz invariant and is thus incompatible with special relativity. A generalisation of the relativistic dispersion relation, using Hamiltonian and momentum operators, leads to the Klein-Gordon equation [7, 8].[1] However, the Klein-Gordon equation is a second-order partial differential equation and subsequently does not uniquely determine the time-evolution of the wavefunction. Alternatively, by finding the half-iterate of the d'Alembert operator,[2] the Klein-Gordon equation simplifies to a first-order partial differential equation; the Dirac equation [9]

$$(i\hbar\gamma^\mu\partial_\mu - mc)\psi = 0, \tag{2.1}$$

written using the Einstein summation convention with a mostly negative metric signature where γ^μ are the Dirac gamma matrices, m is the mass of the Dirac spinor ψ, and c is the speed of light. The Dirac equation is the Euler-Lagrange equation associated to the Dirac Lagrangian density

$$\mathcal{L}_D = \bar{\psi}(i\hbar c\gamma^\mu\partial_\mu - mc^2)\psi, \tag{2.2}$$

where the adjoint spinor is defined by $\bar{\psi} = \psi^\dagger\gamma^0$(cf. [10] for a review on symmetries analytical mechanics). This Lagrangian admits a global U(1) symmetry called the vector symmetry

$$\psi \mapsto e^{i\vartheta}\psi, \quad \bar{\psi} \mapsto \bar{\psi}e^{-i\vartheta}; \quad \vartheta \in \mathbb{R}. \tag{2.3}$$

This symmetry corresponds to a conserved current

$$j^\mu = \bar{\psi}\gamma^\mu\psi, \tag{2.4}$$

called the vector current, via Noether's first theorem [11]. This vector current is the probability four-current of the Dirac fermion. Moreover, the vector current is equivalent to the electromagnetic four-current J^μ;[3] thus, the conservation of probability is equivalent to the conservation of electric charge

$$\partial_\mu j^\mu = 0 \iff \partial_t\rho + \boldsymbol{\nabla}\cdot\boldsymbol{j} = 0, \tag{2.5}$$

where ρ and $\boldsymbol{j}$ are the electric charge and current densities respectively. Using the unitary transformation matrix $U = \frac{1}{\sqrt{2}}(1+\gamma^5\gamma^0)$, the Dirac Lagrangian can be written in the Weyl basis

$$\gamma^\mu_w = U\gamma^\mu U^\dagger, \quad \psi_w = U\psi, \tag{2.6}$$

$$\gamma^\mu_w = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}, \quad \psi_w = \begin{pmatrix} \psi_- \\ \psi_+ \end{pmatrix}, \tag{2.7}$$

where σ^μ is the Pauli four-vector, and $\psi_\pm$ are right-handed and left-handed Weyl spinors respectively [12]. Although the Dirac Lagrangian is invariant under this transformation, it

[1] $(\Box + \mu^2)\psi = 0; \quad \mu \coloneqq \frac{mc}{\hbar}$.
[2] $\Box \coloneqq \frac{1}{c^2}\partial_t^2 - \nabla^2$.
[3] $-ej^\mu = J^\mu$

is easier to see in the Weyl basis that when $m = 0$ the Dirac Lagrangian admits another U(1) symmetry

$$\psi_- \mapsto e^{i\vartheta}\psi_-, \quad \psi_+ \mapsto e^{-i\vartheta}\psi_+ \iff \psi_w \mapsto e^{i\vartheta\gamma_w^5}\psi_w; \quad \vartheta \in \mathbb{R}, \tag{2.8}$$

called the axial symmetry, which has an associated Noether current

$$j_5^\mu = \bar{\psi}\gamma^\mu\gamma^5\psi, \quad \partial_\mu j_5^\mu = 2imc\bar{\psi}\gamma^5\psi, \tag{2.9}$$

known as the axial current. Furthermore, the massless Dirac equation simplifies to two independent Weyl equations [12]

$$i\hbar\sigma^\mu\partial_\mu\psi_+ = 0, \quad i\hbar\bar{\sigma}^\mu\partial_\mu\psi_- = 0. \tag{2.10}$$

Chirality is defined by γ^5; the eigenfunctions of γ^5 have eigenvalues ± 1,[4] and are defined as having positive or negative chirality respectively. The Weyl spinors $\psi_\pm$ are thus the positive and negative chirality components of the Dirac spinor. In the presence of an electromagnetic four-potential A_μ, which we assume to be a non-dynamical background field so that the Maxwell term of the Lagrangian for quantum electrodynamics (QED) can be ignored, the Dirac Lagrangian becomes

$$\mathcal{L} = \bar{\psi}\left(\gamma^\mu\left\{i\hbar c\partial_\mu + ecA_\mu\right\} - mc^2\right)\psi, \tag{2.11}$$

where e is the elementary charge.[5] In matrix form (2.11) is

$$\mathcal{L} = \psi_w^\dagger \begin{pmatrix} c\bar{\sigma}^\mu\widehat{P}_\mu & -mc^2 \\ -mc^2 & c\sigma^\mu\widehat{P}_\mu \end{pmatrix} \psi_w, \tag{2.12}$$

and in the massless case, the equations of motion are

$$\sigma^\mu\widehat{P}_\mu\psi_+ = 0, \quad \bar{\sigma}^\mu\widehat{P}_\mu\psi_- = 0, \tag{2.13}$$

where the four-momentum operator $\widehat{P}_\mu = i\hbar\partial_\mu + eA_\mu$ is given by minimal coupling. We also notice that

$$\sigma^\mu\widehat{P}_\mu = \frac{1}{c}\left(i\hbar\partial_t + e\phi\right)\mathbb{I}_2 + \boldsymbol{\sigma}\cdot\left(i\hbar\boldsymbol{\nabla} + e\boldsymbol{A}\right), \tag{2.14}$$

where ϕ and $\boldsymbol{A}$ are the electric scalar and magnetic vector potentials respectively, $\boldsymbol{\sigma}$ is the Pauli vector, and $\mathbb{I}_2$ is the 2×2 identity matrix. By using the definition of four-momentum,[6] we have the Hamiltonian for Weyl fermions

$$\widehat{H} = -\chi c\boldsymbol{\sigma}\cdot\left(i\hbar\boldsymbol{\nabla} + e\boldsymbol{A}\right) = \chi c\boldsymbol{\sigma}\cdot\hat{\boldsymbol{p}}, \tag{2.15}$$

where $\hat{\boldsymbol{p}}$ is the momentum operator, and $\chi = \pm 1$ for right-handed and left-handed Weyl spinors respectively.

[4] n.b. $\left(\gamma^5\right)^2 = \mathbb{I}_4$.
[5] Here the Dirac spinor has electric charge -e.
[6] $P_\mu = (\frac{1}{c}E, -\boldsymbol{p})$.

2.2 Electronic structure and Berry phase

Insulators are said to be conventional (topologically trivial) if a continuous transformation (homeomorphism) of lattice parameters can transform their band structure into the band structure of the vacuum; the topology is the same as the 'atomic limit'. However, not all insulators are topologically trivial. Suppose a system has an initial Hamiltonian, with a discrete non-degenerate energy spectrum, that evolves continuously as $\widehat{H}(\lambda^\mu)$, where $\lambda^\mu(t)$ are m time-dependent parameters. Furthermore, let $|n(\lambda^\mu(t))\rangle$ be instantaneous eigenstates corresponding to energies $E_n(t)$. Assuming that the Hamiltonian evolves slowly so that $\langle m|\partial_t \widehat{H}|n\rangle \approx 0$ (the adiabatic approximation), then an initial eigenstate evolves as

$$|n(t=t_0)\rangle \xrightarrow{t\to t_1} e^{i\theta_n(t_1)+i\phi_n(t_1)}\,|n(t_0)\rangle\,, \tag{2.16}$$

where $\theta_n(t)$ and $\phi_n(t)$ are the dynamical and Berry phases respectively

$$\theta_n(t_1) = -\frac{1}{\hbar}\int_{t_0}^{t_1} E_n(\tau)\,d\tau, \tag{2.17}$$

$$\phi_n(t_1) = i\int_{t_0}^{t_1} \langle n|\partial_\mu|n\rangle\,\frac{d\lambda^\mu}{d\tau}\,d\tau = i\int_C \langle n|\partial_\mu|n\rangle\,d\lambda^\mu; \qquad \partial_\mu := \frac{\partial}{\partial\lambda^\mu}, \tag{2.18}$$

where C is a rectifiable curve in λ-space. This is the adiabatic theorem [13, 14], and (2.18) describes how the phase of eigenstates changes as the system evolves through parameter space. We now define the Berry potential/connection as

$$A^{\boldsymbol{\lambda}}_\mu = i\,\langle n|\partial_\mu|n\rangle\,. \tag{2.19}$$

In quantum mechanics, the phase of the eigenstates is unphysical, the observable quantities are phase differences; hence, changing the phase of $|n\rangle$ should leave the equations of motion unchanged. Subsequently, the system is invariant under the transformation

$$|n\rangle \to e^{i\beta(\lambda)}\,|n(\lambda)\rangle \implies A^{\boldsymbol{\lambda}}_\mu \to A^{\boldsymbol{\lambda}}_\mu - \partial_\mu\beta(\lambda); \quad \beta(\lambda^\mu) \in C^2(\mathbb{R}^m), \tag{2.20}$$

This redundancy is similar to gauge freedom in electromagnetism and leaves the Berry phase invariant if the path through parameter space is a closed path. Hence, we can construct a U(1) gauge theory for cyclic variations of $\widehat{H}(\lambda^\mu)$, analogous to electromagnetism, where the Berry potential is the gauge connection and the corresponding field strength

$$\Omega^{\boldsymbol{\lambda}}_{\mu\nu} = \partial_\mu A^{\boldsymbol{\lambda}}_\nu - \partial_\nu A^{\boldsymbol{\lambda}}_\mu, \tag{2.21}$$

is called the Berry curvature. In three dimensions, we also define the Berry curvature pseudovector

$$\boldsymbol{\Omega}^{\boldsymbol{\lambda}} = \boldsymbol{\nabla}_{\boldsymbol{\lambda}} \times \boldsymbol{A}^{\boldsymbol{\lambda}}. \tag{2.22}$$

Notice that the pseudovector is divergence-free because it is a curl, so the 'Berry flux' is conserved when the Berry potential is non-singular. Moreover, the antisymmetric tensor and pseudovector are related via

$$\Omega^{\boldsymbol{\lambda}}_{\mu\nu} = \epsilon_{\mu\nu\rho}\Omega^{\boldsymbol{\lambda}}_\rho, \tag{2.23}$$

where $\epsilon_{\mu\nu\rho}$ is the Levi-Civita symbol. The Berry curvature is a gauge invariant quantity thus, using Stokes' theorem [15], we can construct another expression for the Berry phase

$$\phi_n = \oint_C A_\mu^{\boldsymbol{\lambda}}\, d\lambda^\mu = \int_S \Omega_{\mu\nu}^{\boldsymbol{\lambda}}\, ds^\mu \wedge ds^\nu, \tag{2.24}$$

where C is the boundary of the surface S in parameter space. The right-hand side is the integral of the curvature form of a vector bundle on a smooth manifold hence by using Chern-Weil theory we have that

$$\phi_n = 2\pi c_1, \tag{2.25}$$

where c_1 is the first Chern class which is a topologically invariant integer [16], also known as the Thouless-Kohmoto-Nightingale-Nijs (TKNN) invariant because of its relation to the quantum Hall effect [17]. Bloch's theorem states that the energy eigenstates of the n^{th} band of a perfect lattice are of the form

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{r}} u_{n\boldsymbol{k}}(\boldsymbol{r}), \tag{2.26}$$

where $\hbar\boldsymbol{k}$ is crystal momentum, and the Bloch functions $u_{n\boldsymbol{k}}(\boldsymbol{r})$ have the periodicity of the direct lattice [18]. Thus, for a crystal lattice the Berry potential becomes

$$\boldsymbol{A}_n^{\boldsymbol{k}} = i \left\langle u_{n\boldsymbol{k}} \middle| \boldsymbol{\nabla}_{\boldsymbol{k}} \middle| u_{n\boldsymbol{k}} \right\rangle, \tag{2.27}$$

where n is now the band index, and we have that

$$\int \Omega_{\mu\nu}^{n\boldsymbol{k}}\, ds^\mu \wedge ds^\nu = 2\pi c, \tag{2.28}$$

is a a topological invariant, where c is called the Chern number. For 2D crystals (2.28) can be an integral over the Brillouin zone, and we have a Chern number associated to each band. The electronic bands of topologically trivial insulators have a Chern number of 0, and the most famous Chern insulator (material characterised by Chern numbers) is the Haldane model of graphene [19]. At the interface between topological and conventional insulators the different topological invariants force band degeneracy at the boundary and the adiabatic theorem does not apply. Subsequently, 2D topological insulators have topologically protected metallic surface states because the vacuum is topologically trivial. Thus far, we have seen Chern numbers used to classify two dimensional topological insulators but they can also characterise three dimensional materials ; specifically, Dirac and Weyl semimetals.

3 Dirac and Weyl semimetals

Weyl semimetals (WSMs) are materials possessing Weyl fermions as their quasiparticle excitations. Specifically, WSMs are zero band gap semimetals where the effective Hamiltonian near band degeneracies, called Weyl points, takes the form

$$\widehat{H}(\boldsymbol{k}) = \varepsilon_0 \mathbb{I}_2 + \chi\hbar v_f \boldsymbol{\sigma}\cdot(\boldsymbol{k} - \boldsymbol{k}^w), \tag{3.1}$$

where ε_0 and $\hbar \boldsymbol{k}^w$ are the energy and crystal momentum at the Weyl point respectively,[7] v_f is the Fermi velocity, and $\chi = \pm 1$ is the Weyl point's chirality [2]. When ϵ_0 is approximately the Fermi energy the quasiparticle excitations are Weyl fermions because the equations of motion of the WSM Hamiltonian are the same as (2.15). In addition, the von Neumann-Wigner theorem ensures that Weyl points are isolated in the Brillouin zone creating 'Weyl cones' at band crossings [20]. By direct computation, and assuming $\boldsymbol{k}^w = 0$ for now, we find that the eigenstates of the WSM Hamiltonian are

$$v_\pm = \begin{pmatrix} k_3 \pm |\boldsymbol{k}| \\ k_1 + ik_2 \end{pmatrix}; \quad \varepsilon_\pm = \varepsilon_0 \pm \chi \hbar v_f |\boldsymbol{k}|, \tag{3.2}$$

with energies $\varepsilon_\pm$ respectively. Using (2.22), this corresponds to a Berry curvature pseudovector of

$$\boldsymbol{\Omega}^{\boldsymbol{k}} = \frac{\chi(\boldsymbol{k} - \boldsymbol{k}^w)}{2|\boldsymbol{k} - \boldsymbol{k}^w|^3}. \tag{3.3}$$

Thus, using the divergence theorem [21] we can see that Weyl points are monopoles of Berry flux and have Chern number χ

$$\oint_S \Omega^{\boldsymbol{k}}_{\mu\nu}\, ds^\mu \wedge ds^\nu = 2\pi\chi, \tag{3.4}$$

where the surface S encloses a Weyl point at $\boldsymbol{k}^w$ [5]. Thus, Weyl points are topologically protected band crossings because a homeomorphism cannot remove them without changing the Chern number (unless two Weyl points are brought together). Alternatively, Weyl nodes can be viewed as possessing a 'topological charge' χ, and the Nielsen-Ninomiya theorem, which shows that Weyl points come in pairs of opposite chirality, represents the conservation of this topological charge [22]. Crystals with inversion symmetry[8] have symmetric Berry curvatures ($\boldsymbol{\Omega}^{\boldsymbol{k}} = \boldsymbol{\Omega}^{-\boldsymbol{k}}$). However, when a crystal has time-reversal symmetry ($|u_{-\boldsymbol{k}}\rangle = |u_{\boldsymbol{k}}\rangle^\dagger$) the Berry potential is

$$A_\mu = \langle u^\dagger_{-k} | \partial_\mu u^\dagger_{-k} \rangle = \langle \partial_\mu u_{-k} | u_{-k} \rangle, \tag{3.5}$$

and the Berry curvature is antisymmetric

$$\begin{aligned} \Omega^{\boldsymbol{k}}_{\mu\nu} &= \partial_\mu \langle \partial_\nu u_{-k} | u_{-k} \rangle - \partial_\nu \langle \partial_\mu u_{-k} | u_{-k} \rangle \\ &= \langle \partial_\nu u_{-k} | \partial_\mu u_{-k} \rangle - \langle \partial_\mu u_{-k} | \partial_\nu u_{-k} \rangle = -\Omega^{-\boldsymbol{k}}_{\mu\nu}. \end{aligned} \tag{3.6}$$

Subsequently, a WSM can have either time-reversal symmetry, in which case Weyl nodes at opposite points of the Brillouin zone have the same chirality, or inversion symmetry which produces Weyl points of the same chirality at opposite points. Moreover, because the total chirality is 0, the number of Weyl points in a noncentrosymmetric WSM must be a multiple of 4 [2]. A WSM is forbidden from having both symmetries due to Kramers' degeneracy, eigenstates related via time-reversal are degenerate in time-reversal symmetric

[7] ϵ_0 is near the Fermi energy.
[8] i.e. $\boldsymbol{k} \equiv -\boldsymbol{k}$.

systems with half-integer spin [23]. Hence, when a crystal possess both symmetries we have that the energy obeys

$$E_{\uparrow}(\boldsymbol{k}) = E_{\downarrow}(-\boldsymbol{k}) = E_{\downarrow}(\boldsymbol{k}), \tag{3.7}$$

each band is at least doubly degenerate across the Brillouin zone because the energy is invariant under spin reversal, which is not true for (3.1). In this case, the Hamiltonian with the required degeneracies is

$$\widehat{H}(\boldsymbol{k}) = \varepsilon_0 \mathbb{I}_4 + \hbar v_f \begin{pmatrix} -\boldsymbol{\sigma}\cdot(\boldsymbol{k}-\boldsymbol{k}^d) & 0 \\ 0 & \boldsymbol{\sigma}\cdot(\boldsymbol{k}-\boldsymbol{k}^d) \end{pmatrix}, \tag{3.8}$$

where $\mathbb{I}_4$ is the $4{\times}4$ identity matrix, and there is a fourfold band crossing at energy ε_0 at $\boldsymbol{k}^d$ in the Brillouin zone. This is identical to the Hamiltonian of massless Dirac fermions (2.12); hence, it is the Hamiltonian of the Dirac semimetal (DSM) whose quasiparticle excitations are massless Dirac fermions [24]. Subsequently, we call the degeneracy at $\boldsymbol{k}^d$ a Dirac point. Two Weyl points of opposite chirality coincide at $\boldsymbol{k}^d$ so the Dirac point has no overall Chern number and is not topologically protected [2]. However, when two Weyl points meet they may annihilate each other introducing mass terms into the Hamiltonian similar to (2.12), and creating a band gap at $\boldsymbol{k}^d$. Consequently, DSMs often have additional symmetries which necessitate a band crossing, creating stable Dirac points.

3.1 Magnetic WSMs

The first hypothesised WSMs were centrosymmetric materials with magnetic ordering breaking time-reversal symmetries, often called magnetic WSMs. In particular, *ab initio* calculations predicted that the pyrochlore iridates $A_2Ir_2O_7$ (A = yttrium or a lanthanide) possess an all-in/all-out magnetic structure, where all Ir magnetic moments on a tetrahedron's vertices point either inward or outward, breaking time-reversal symmetry [25]. Moreover, pyrochlore iridates were hypothesised to have multiple electronic phases dependent on the electronic correlation energy U, becoming WSMs with 24 Weyl points at an intermediate $U \sim 1.5\,\mathrm{eV}$, and Mott insulators at $U > 2\,\mathrm{eV}$. Experiments have confirmed the existence of magnetic order in pyrochlore iridates and their all-in/all-out structure [26]. However, the WSM phase has yet to be discovered, although some pyrochlore iridates remain promising WSM candidates. In particular, the large hall conductivity of $Nd_2Ir_2O_7$ and its low magnetisation, due to antiferromagnetic all-in/all-out order, suggests that the material's spontaneous hall effect has a topological origin [27].

3.2 Noncentrosymmetric WSMs

Using first principles calculations, TaAs and similar noncentrosymmetric body-centered-tetragonal crystals were predicted to have 24 Weyl points across their Brillouin zones and topologically protected surface states called Fermi arcs [28]. The prediction of such materials was particularly important because previous theoretical WSMs were difficult to synthesise magnetic materials. The Berry flux through an open surface between two Weyl points of opposite chirality is non-zero; thus, each 2D plane between Weyl points is a 2D Chern insulator that must have metallic edge states [29]. These gapless edge states on the Fermi surface of the WSM boundary are called Fermi arcs and can be observed using angle-resolved photoemission spectroscopy (ARPES). Moreover, such metallic states must terminate because planes beyond Weyl points are conventional insulators; the ends of Fermi arcs are projections of Weyl points onto the Fermi surface. ARPES employs the photoelectric effect to measure crystal momenta, using an ultraviolet (UV) laser incident on a sample to produce photoelectrons which are analysed in a spectrometer [30]. Because the electron momentum parallel to the crystal surface is unaffected by the workfunction (conserved during photoemission) ARPES is a useful probe of crystal momenta in 2D materials[9]. If the material workfunction ϕ is known, the binding energy

$$E_b = \hbar\nu - \phi - E_k, \tag{3.9}$$

may be inferred, where ν is the laser frequency and E_k is the photoelectron kinetic energy measured by an electron spectrometer. Using the crystal momentum parallel to the surface

$$\hbar k_{||} = \sqrt{2m_e E_k}\sin\theta, \tag{3.10}$$

where m_e is the electron mass and θ is the angle between the photoemission direction and surface normal, the Fermi surface ($E_b = 0$) of a 2D material can be mapped. ARPES experiments have identified the characteristic Fermi arcs of a WSM in TaAs with *ab initio* band structure predictions closely matching observation [31].

3.3 Dirac matter

Band structure calculations predict that Na_3Bi and Cd_3As_2 are DSMs, each possessing a pair of symmetry protected Dirac points and Fermi arcs despite no separation between the Weyl points [32, 33]. Na_3Bi forms hexagonal crystals in the $P6_3/mmc$ space group with threefold rotational symmetry which forbids the existence of a band gap; potential Dirac points are protected by C_3 symmetry. Hence, the introduction of a uniaxial strain or an external magnetic field could destroy the Dirac points via removal of C_3 symmetry. Contrastingly, Cd_3As_2 exhibits temperature dependent polymorphism, possessing a low-temperature ($< 750\,\mathrm{K}$) body-centered-tetragonal $I4_1/acd$ phase, an intermediate-temperature (750–875 K) primitive tetragonal $P4_2/nmc$ phase, and a high-temperature ($> 875\,\mathrm{K}$) face-centered-cubic $Fm\bar{3}m$ phase [34, 35].

Initial *ab initio* electronic structure calculations for Cd_3As_2 were preformed erroneously, using the noncentrosymmetric space group $I4_1cd$ as the low-temperature polymorph, and subsequently predicted a low-temperature WSM phase [33]. However, the actual low-temperature centrosymmetric $I4_1/acd$ phase has fourfold rotational symmetry, like the intermediate-temperature phase, which prevents the opening of a band gap at the hypothesised Dirac points [34, 36]. ARPES measurements of both Na_3Bi and Cd_3As_2 (in the intermediate-temperature polymorph) monocrystals show linear dispersion at potential Dirac points consistent with the presence of Dirac cones within the Brillouin zone [37, 38].

[9] n.b. the momentum of a UV photon is negligible

4 The chiral anomaly

The central mathematical object used to investigate quantum field theories (QFTs) in the path integral formalism is the partition function [39]

$$Z[J] = \int \mathcal{D}\phi \, \exp\left[\frac{i}{\hbar}S[\phi] + \frac{i}{\hbar}\int d^4x \, J(x)\phi(x)\right], \tag{4.1}$$

where $S[\phi]$ is the classical action of the dynamical field $\phi(x)$ which is coupled to a source field $J(x)$. In particular, the partition function can be used to express expectation values in a QFT

$$\frac{1}{Z[0]}\int \mathcal{D}\phi \, F(x) e^{i\frac{S[\phi]}{\hbar}} = \langle F(x)\rangle \, . \tag{4.2}$$

To calculate path integrals we must perform a wick rotation (imaginary time substitution) so that the classical action is an integral over Euclidean spacetime,[10] after which we can transform the time axis back to $\mathbb{R}$ via analytic continuation. Anomalies of a QFT are symmetries of the classical action which are not preserved when the theory is quantised; symmetries violated by quantum processes. In particular, transformations which are symmetries of the classical action but not symmetries of the partition function are anomalous. Because of Noether's first theorem this manifests as the non-conservation of a classically conserved quantity. Moreover, the Slavnov–Taylor identities [40] show that, for symmetries linear in the fields (such as the vector and axial symmetries), a transformation in the path integral 'measure' of the form

$$\mathcal{D}\phi \mapsto \mathcal{D}\phi \, \det[\mathcal{J}], \tag{4.3}$$

where $\mathcal{J}$ is a singular Jacobian matrix, is necessary and sufficient for a symmetry to be anomalous; anomalies are characterised by singular path integral Jacobians for linear symmetries. For example, the axial symmetry is famously violated by triangle Feynman diagrams, resulting in the neutral pion's high decay rate known as the Adler-Bell-Jackiw (ABJ) anomaly [3, 4]. As we shall see, this anomaly (known as the chiral anomaly outside of particle physics) can be viewed as a topological invariant of a QFT (specifically QED) via the Atiyah-Singer index theorem (cf. [10] for a review) [6].

4.1 Quantum field theory

Because Dirac spinors are anticommuting Grassmann variables, the fermionic path integral is a Berezin integral [41]. Consequently, to write the path integral for Dirac fermions we must first decompose the Dirac spinor and its adjoint using an orthonormal basis $\{\phi_n(x)\}$, of eigenfunctions of the Dirac operator $\not{D}$,[11] and the independent Grassmann variables θ_n and $\bar{\xi}_m$

$$\psi = \sum_n \theta_n \phi_n(x) = \sum_n \theta_n \langle x|n\rangle, \tag{4.4}$$

$$\bar{\psi} = \sum_m \bar{\xi}_m \phi_m^\dagger(x) = \sum_m \bar{\xi}_m \langle m|x\rangle. \tag{4.5}$$

[10] Integrals over Minkowski spacetime are much harder to manipulate.

[11] Hereafter we use Feynman slash notation and $i\hbar\widehat{D}_\mu = \widehat{P}_\mu$.

Subsequently, the path integral 'measure' can be seen as a Berezin integral 'measure' under the coordinate transformation

$$\theta \mapsto \theta_n \langle x|n\rangle\,, \quad \bar{\xi} \mapsto \bar{\xi}_m \langle m|x\rangle\,, \tag{4.6}$$

$$\mathcal{D}\psi\mathcal{D}\bar{\psi} = \left[\det \langle x|n\rangle \det \langle m|x\rangle\right]^{-1} \lim_{\substack{N\to\infty\\ M\to\infty}} \left\{\prod_n^N d\theta_n \prod_m^M d\bar{\xi}_m\right\} = \lim_{N\to\infty} \prod_n^N d\theta_n d\bar{\xi}_n. \tag{4.7}$$

Under the local vector symmetry the Dirac spinor becomes

$$\psi' = \sum_n \theta'_n \phi_n(x); \quad \theta'_n = e^{i\vartheta(x)}\theta_n, \tag{4.8}$$

and similarly for the adjoint spinor. Subsequently, the path integral measure transformation is

$$\mathcal{D}\psi\mathcal{D}\bar{\psi} \mapsto \left[\det\left(e^{i\vartheta(x)\mathbb{I}}\right)\det\left(e^{-i\vartheta(x)\mathbb{I}}\right)\right]^{-1}\mathcal{D}\psi\mathcal{D}\bar{\psi}, \tag{4.9}$$

where $\mathbb{I}$ is the identity matrix. Clearly the path integral measure is invariant under the vector symmetry; thus, by the Slavnov–Taylor identities, the vector symmetry is retained by the quantum theory and electric charge is conserved. Contrastingly, this is not the case for the axial transformation; the axial symmetry is anomalous. The following is an outline of Fujikawa's proof of the existence of the chiral anomaly [42, 43]. Under the local axial symmetry, the Grassmann variables transform as

$$\theta'_n = \sum_m C_{nm}\theta_m, \quad \bar{\xi}'_n = \sum_m C_{nm}\bar{\xi}_m, \tag{4.10}$$

$$C_{nm} = \delta_{nm} + i\int d^4x\, \vartheta(x)\phi_n^\dagger(x)\gamma^5\phi_m(x). \tag{4.11}$$

Using the identities

$$\ln\left(\mathbb{I} + \epsilon A\right) = \epsilon A + \mathcal{O}(\epsilon^2), \tag{4.12}$$

$$\det\left(e^A\right) = e^{\mathrm{Tr}(A)}, \tag{4.13}$$

one can see that the path integral measure changes by the Jacobian

$$\begin{aligned}\mathcal{J}[\vartheta] &= (\det C)^{-2} = \exp\left[-2\mathrm{Tr}(\ln C)\right]\\ &= \exp\left[-2\mathrm{Tr}\left(i\int d^4x\,\vartheta(x)\phi_n^\dagger(x)\gamma^5\phi_m(x)\right)\right]\\ &= \exp\left[-2i\int d^4x\,\vartheta(x)\sum_n\phi_n^\dagger(x)\gamma^5\phi_n(x)\right].\end{aligned} \tag{4.14}$$

The sum appearing in the Jacobian is clearly divergent (by orthonormality) and requires the introduction of a regulator, or ultraviolet cut-off, to evaluate the integral (cf. Fujikawa's method [42]). However, we will explore a more elegant topological approach that utilises topological and analytical indices, after demonstrating the relationship between the chiral anomaly and the axial current. Henceforth, we define the axial anomaly,[12] $\mathcal{A}$, using the Jacobian of the path integral measure

$$\mathcal{J}[\vartheta] = \exp\left[-\int d^4x\, \vartheta(x)\mathcal{A}\right]. \tag{4.15}$$

Under the axial transformation in Euclidean spacetime the partition function becomes

$$Z'[A] = \int \mathcal{D}\psi'\mathcal{D}\bar{\psi}' \exp\left[\frac{1}{\hbar}\int d^4x\; -\frac{1}{4\mu_0 c}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}'(i\hbar\not{D} - mc)\psi'\right], \tag{4.16}$$

where $F_{\mu\nu}$ is the electromagnetic field tensor, and μ_0 is the vacuum permeability. This is equivalent to a relabeling of the path integration variables so the partition function must be invariant. Considering the change in the QED Lagrangian $\mathcal{L}_{\text{QED}}$, we have

$$Z[A] = \int \mathcal{D}\psi\mathcal{D}\bar{\psi}\, \mathcal{J}[\vartheta] \exp\left[\frac{1}{\hbar c}\int d^4x\, \mathcal{L}_{\text{QED}} + c\vartheta(x)(\partial_\mu j_5^\mu - 2imc\bar{\psi}\gamma^5\psi)\right]. \tag{4.17}$$

Thus, to first-order in $\vartheta(x)$ we have

$$Z[A] = \int \mathcal{D}\psi\mathcal{D}\bar{\psi} \left\{1 + \frac{1}{\hbar}\int d^4x\, \vartheta(x)(\partial_\mu j_5^\mu - 2imc\bar{\psi}\gamma^5\psi - \mathcal{A})\right\} \exp\left[\frac{1}{\hbar c}\int d^4x\, \mathcal{L}_{\text{QED}}\right]. \tag{4.18}$$

Subsequently, by the fundamental lemma of the calculus of variations [44] we have the anomalous divergence for the axial current

$$\partial_\mu j_5^\mu = 2imc\bar{\psi}\gamma^5\psi + \mathcal{A}, \tag{4.19}$$

which we call the chiral anomaly. The Dirac operator, $\not{D}$, is Hermitian in Euclidean space. If ϕ_n is an eigenfunction of $\not{D}$, with a non-zero eigenvalue λ_n, then $\gamma^5\phi_n$ is also an eigenfunction

$$\not{D}(\gamma^5\phi_n) = -\gamma^5\not{D}(\phi_n) = -\lambda_n\gamma^5\phi_n. \tag{4.20}$$

Moreover, Hermiticity of $\not{D}$ implies that ϕ_n and $\gamma^5\phi_n$ are orthogonal (as they have different eigenvalues) because of the spectral theorem [45]. When the eigenvalue vanishes ϕ_n and $\gamma^5\phi_n$ are called zero modes of the Dirac operator. In addition, γ^5 is Hermitian; hence, the spectral theorem implies there exists an orthonormal basis that diagonalises γ^5 and spans $\text{Ker}(\not{D})$. Ignoring the infinitesimal parameter $\vartheta(x)$ for now, only the zero modes contribute to the integral in (4.14) because of orthonormality, and we have that

$$\int d^4x \sum_n \phi_n^\dagger\gamma^5\phi_n = \sum_n \int d^4x\, \phi_{n+}^{0\dagger}\phi_{n+}^0 - \sum_n \int d^4x\, \phi_{n-}^{0\dagger}\phi_{n-}^0 = n_+ - n_-, \tag{4.21}$$

$$\int d^4x\, \mathcal{A} = 2i(n_+ - n_-), \tag{4.22}$$

where $n_\pm$ are the number of positive and negative chirality zero modes, $\phi_{n\pm}^0$, respectively. This is precisely the analytical index of the Dirac operator projected onto the positive chirality subspace

$$\text{index}(\not{D}_+) = n_+ - n_-, \tag{4.23}$$

[12] In this work the axial anomaly is a quantity whereas the chiral anomaly is an equation.

$$\not{D}_\pm \coloneqq \not{D} P_\pm = \not{D}\big|_{\{\pm\}}, \tag{4.24}$$

where $P_\pm = \frac{1}{2}(1 \pm \gamma^5)$ is the chirality projection operator. By the Atiyah–Singer index theorem, this index coincides with the Dirac operator's topological index on a compact manifold [6]. This implies that the chiral anomaly is a topological invariant of quantum electrodynamics and no regularisation procedure can restore the axial symmetry. On a four dimensional compact manifold Ω, with no boundary and Riemannian curvature R, the topological index is [46, 47]

$$\text{index}(\not{D}_+) = \int_\Omega \hat{A}(\Omega)\text{ch}(F). \tag{4.25}$$

The integrand, called the index density, only includes 4-form terms and the curvature form is given by

$$F = \frac{e}{\hbar}\, dA = \frac{e}{2\hbar} F_{\mu\nu}\, dx^\mu \wedge dx^\nu, \tag{4.26}$$

where dA is the exterior derivative of the four-potential. The characteristic classes $\text{ch}(F)$ and $\hat{A}(\Omega)$ are the Chern character of the curvature form, and Dirac genus of the manifold respectively

$$\text{ch}(F) = \text{Tr}\left(\exp\left[\frac{i}{2\pi}F\right]\right), \tag{4.27}$$

$$\hat{A}(\Omega) = \sqrt{\det\left(\frac{\frac{i}{4\pi}R}{\sinh\frac{i}{4\pi}R}\right)}. \tag{4.28}$$

Now we choose the manifold $S^4 = \Omega$ so the Riemannian curvature is constant and $\hat{A}(\Omega) = 1$. Thus the index becomes

$$\text{index}(\not{D}_+) = \int_{S^4} \text{ch}(F) = \frac{1}{2!}\left(\frac{i}{2\pi}\right)^2 \int_{S^4} \text{Tr}(F \wedge F). \tag{4.29}$$

Assuming the action due to the Maxwell Lagrangian converges

$$\int_{\mathbb{R}^4} d^4x\, F_{\mu\nu}F^{\mu\nu} < \infty \iff A_\mu \xrightarrow{|x|\to\infty} \frac{\hbar}{e}\partial_\mu\vartheta; \quad \vartheta(x) \in C^2(\mathbb{R}^4), \tag{4.30}$$

we can stereographically project our problem from S^4 to $\mathbb{R}^4$. Hence $\text{ch}(F)$ coincides with the Dirac operator's index density in Euclidean space if the gauge fields are asymptotically 'flat' (A_μ is a pure gauge at infinity) [48, 49]. Using antisymmetry of the wedge product, we identify the chiral anomaly in Euclidean spacetime as[13]

$$\partial_\mu j_5^\mu - 2imc\bar{\psi}\gamma^5\psi = \mathcal{A}[A_\mu] = \frac{-ie^2}{16\pi^2\hbar^2}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}. \tag{4.31}$$

[13]The $-i$ factor disappears in Minkowski spacetime.

4.2 Condensed matter

Electron transport phenomena in condensed matter systems are effectively described by the semiclassical model of electron dynamics [50]. Our goal is to describe the motion of electrons between electron-ion scattering events however, we cannot apply an electron's classical equations of motion like in the Drude model due to the uncertainty principle; simultaneously definite position and momentum are forbidden. Subsequently, we introduce the n^{th} band electron wave packet in a crystal

$$\psi_n(\boldsymbol{r},t) = \int \frac{d^3\boldsymbol{k}}{(2\pi)^3}\, g(\boldsymbol{k})\psi_{n\boldsymbol{k}} e^{-i\omega_n(\boldsymbol{k})t}; \tag{4.32}$$

$$|\boldsymbol{k}' - \boldsymbol{k}| > \Delta k \implies g(\boldsymbol{k}') \approx 0, \tag{4.33}$$

where $\hbar\omega_n(\boldsymbol{k})$ is the energy of the Bloch wavefunction $\psi_{n\boldsymbol{k}}$, the weight function $g(\boldsymbol{k})$ determines the distribution of the wave packet, and Δk is much smaller than the dimensions of the Brillouin zone. The mean velocity of a Bloch electron[14] is also the group velocity of the wave packet. Thus, we can posit that the wave packet obeys the semiclassical equations of motion

$$\dot{\underset{\sim}{\boldsymbol{r}}} = \boldsymbol{\nabla}_{\boldsymbol{k}}\omega_n(\boldsymbol{k}), \quad \hbar\dot{\underset{\sim}{\boldsymbol{k}}} = -e\boldsymbol{E} - e\dot{\underset{\sim}{\boldsymbol{r}}} \times \boldsymbol{B}, \tag{4.34}$$

where $\underset{\sim}{\boldsymbol{r}}$ and $\hbar\underset{\sim}{\boldsymbol{k}}$ are the wave packet position and crystal momentum respectively, when the spread of the wave packet is much smaller than the wavelength of external electric and magnetic fields ($\boldsymbol{E}$ and $\boldsymbol{B}$). In the presence of a Berry curvature the wave packet velocity shifts by $-\dot{\underset{\sim}{\boldsymbol{k}}} \times \boldsymbol{\Omega}^{\boldsymbol{k}}$ [51]. However, due to deviations from a perfect lattice caused by the thermal vibrations of ions, electron-ion scattering occurs in real crystals and the electron distribution function $f(\underset{\sim}{\boldsymbol{r}}, \underset{\sim}{\boldsymbol{k}}, t)$ does not obey the semiclassical equations of motion. Instead, we have the Boltzmann kinetic equation

$$\frac{\partial f}{\partial t} + \dot{\underset{\sim}{\boldsymbol{r}}} \cdot \boldsymbol{\nabla}_{\underset{\sim}{\boldsymbol{r}}} f + \dot{\underset{\sim}{\boldsymbol{k}}} \cdot \boldsymbol{\nabla}_{\underset{\sim}{\boldsymbol{k}}} f = \left(\frac{\partial f}{\partial t}\right)_{\text{coll}}, \tag{4.35}$$

where the collision term on the right-hand side encodes the effect of scattering on electron transport [52]. Using the semiclassical phenomenological equations of motion and the Boltzmann kinetic equation one can derive the hydrodynamic equation [5]

$$\partial_t N^{\pm} + \boldsymbol{\nabla}_{\boldsymbol{r}} \boldsymbol{j}^{\pm} = \frac{\chi e^2}{4\pi^2\hbar^2 c} \boldsymbol{E} \cdot \boldsymbol{B}, \tag{4.36}$$

where $N^{\pm}$ and $\boldsymbol{j}^{\pm}$ are the electron and current densities at Weyl points of chirality $\chi = \pm 1$ respectively. The sums of the chiral specific electron and current densities are the total electron and electric current densities respectively (N and $\boldsymbol{j}$); however, their differences ($N^+ - N^-$ and $\boldsymbol{j}^+ - \boldsymbol{j}^-$) are the axial charge and 3-current densities respectively

$$\partial_t N + \boldsymbol{\nabla}_{\boldsymbol{r}} \boldsymbol{j} = 0, \tag{4.37}$$

$$\partial_t (N^+ - N^-) + \boldsymbol{\nabla}_{\boldsymbol{r}} (\boldsymbol{j}^+ - \boldsymbol{j}^-) = \frac{e^2}{2\pi^2\hbar^2 c} \boldsymbol{E} \cdot \boldsymbol{B}, \tag{4.38}$$

[14] n.b. $v_n = \boldsymbol{\nabla}_{\boldsymbol{k}}\omega_n$.

([4.37]) represents the conservation of electric charge whereas ([4.38]) is the chiral anomaly.[15] This non-conservation of axial charge is seen as a large electric current between Weyl points of opposite chirality that conserves the overall electric charge of a Weyl semimetal. This produces a large negative longitudinal magnetoresistance (LMR) in WSMs, when external electric and magnetic fields are parallel, that is highly anisotropic due to the $\boldsymbol{E} \cdot \boldsymbol{B}$ factor.

5 Magnetoresistance experiments

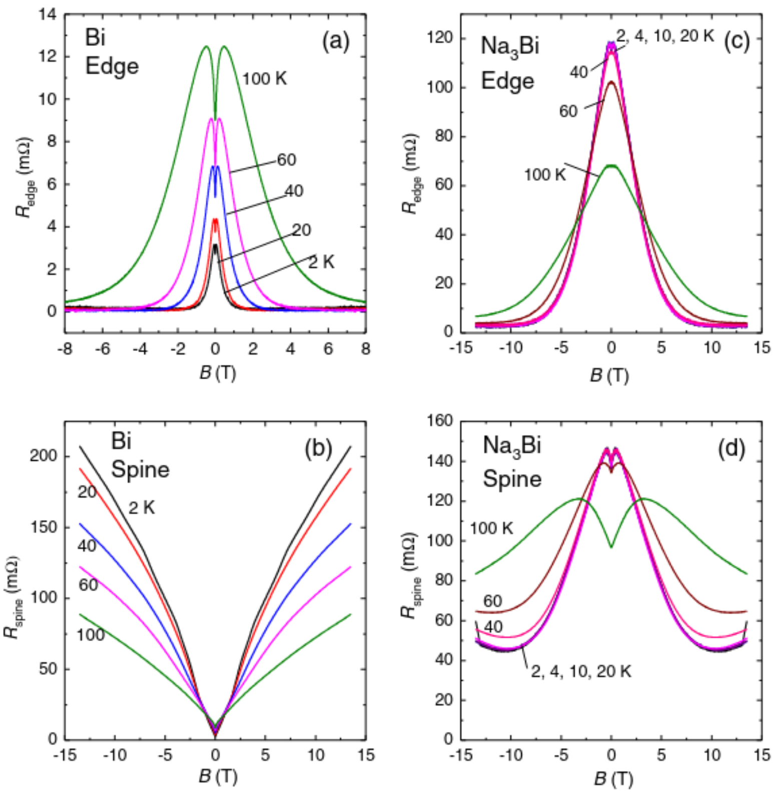

Figure 1. [53]Squeeze test results in pure Bi and Na_3Bi at various temperatures shown on the left and right respectively.

Current jetting artefacts are of high concern when conducting magnetoresistance experiments in semimetals with high carrier mobilities. In the presence of a large magnetic field colinear to the electric current the cyclotronic motion of charge carriers increases the transverse resistance, but the longitudinal resistance is unchanged [53]. Subsequently, the electric current forms a narrow jet in the longitudinal direction between the current contacts on the material, resulting in a low current density at the edges of a sample. Thus, the voltage dropped at the edge of the material is lower than the on-axis potential difference. If the former is detected instead of the latter the reduced voltage would be interpreted as an increased longitudinal conductivity; a negative linear magnetoresistance. This effect is hypothesised to be most pronounced in high-mobility semimetals such as Bi [53, 54].

[15] n.b. $\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma.} = \frac{8}{c}\boldsymbol{E} \cdot \boldsymbol{B}$.

To identify extrinsic negative LMRs caused by current jetting artefacts one employs the 'squeeze test' on semimetal samples, where pairs of voltage contacts are placed along the current contact central axis (along the 'spine') and along the material edge parallel to the central axis. These contacts simultaneously measure the spine and edge voltages (V_{spine} and V_{edge} respectively) from which the effective resistances are calculated ($R_{\text{spine}}(\boldsymbol{B})$ and $R_{\text{edge}}(\boldsymbol{B})$ respectively). The current jetting artefacts produce an increasing $R_{\text{edge}}(\boldsymbol{B})$ and decreasing $R_{\text{spine}}(\boldsymbol{B})$; consequently, opposite trends in the effective resistances is indicative of an extrinsic spurious negative LMR [53]. The opposite trends in the edge and spine effective resistances in Fig.1 for pure Bi suggest that the apparent negative magnetoresistance at the Bi edge is due to current jetting. Contrastingly, the correlation between edge and spine resistances within Na_3Bi, a low-mobility semimetal, implies an intrinsic negative magnetoresistance.

Magnetoresistance experiments have observed a weak negative LMR in TaAs below 75 K with high anisotropy [55]. However, due to the high carrier mobility of TaAs current jetting artefacts arise at $\boldsymbol{B} \sim 0.4\,\text{T}$ which is far below the field strength at which an anomaly induced negative LMR is expected to manifest ($\boldsymbol{B} \sim 7\,\text{T}$) [53]. Consequently, an anomaly induced negative magnetoresistance is impossible to identify in TaAs and previously observed LMRs are likely extrinsic.

Contrastingly, evidence for the chiral anomaly is more promising in DSMs. An external $\boldsymbol{B}$ field breaks the time-reversal symmetry of the DSM Na_3Bi, splitting both Dirac points into Weyl points of opposite chirality [24]. Thus, Na_3Bi exhibits an anisotropic negative magnetoresistance below 100 K [53]. Moreover, the low carrier mobility of Na_3Bi excludes current jetting effects, suggesting that the negative LMR is intrinsic and the strongest evidence for the chiral anomaly in condensed matter.

6 Conclusion

We have seen how Dirac and Weyl semimetals can be classified as topological materials and that this gives rise to topologically protected Fermi arcs on their Fermi surfaces. Similarly, the nature of the chiral anomaly as a topological invariant of quantum electrodynamics has been laid out. Moreover, evidence for existing Dirac and Weyl semimetals has been reviewed, with ARPES providing strong evidence for the WSM candidate TaAs, and the DSM candidates Na_3Bi and Cd_3As_2. Using the semiclassical model of electron dynamics and the Boltzmann kinetic equation, the anomaly-induced negative linear magnetoresistance in WSMs has been derived. Evidence for the chiral anomaly in WSMs like TaAs is poor due to their high carrier mobilities producing current jetting artefacts. The DSM Na_3Bi has a low carrier mobility which precludes current jetting. Hence, the negative linear magnetoresistance observed in Na_3Bi is the best evidence for the chiral anomaly in condensed matter systems to date.

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7 Lay summary

The elementary particles of the universe, which are the building blocks of all matter, come in mirror-image pairs; electrons can either be left-handed or right-handed. This handedness property is called chirality, and under normal circumstances, the number of left-handed and right-handed particles remains constant. Additionally, electrons obey the conservation of mass; the total number of electrons is always the same. Some hypothetical fundamental particles, specifically Weyl fermions, are predicted to violate this conservation law in a process called the chiral anomaly. In particular, some Weyl fermions are only left-handed while others are exclusively right-handed. As a result, the total number of Weyl fermions is not necessarily constant; instead, the difference between the number of left and right-handed Weyl fermions is conserved. Although actual Weyl fermions have not been discovered in nature, electrons behave as if they were Weyl fermions in materials called Weyl semimetals. This leads to an imbalance in the numbers of left and right-handed electrons depending on the strength of electric and magnetic fields applied to Weyl semimetals.